

# Adaptive Law-Based Transformation *altx*

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Data and Compute Intensive Sciences Research Group

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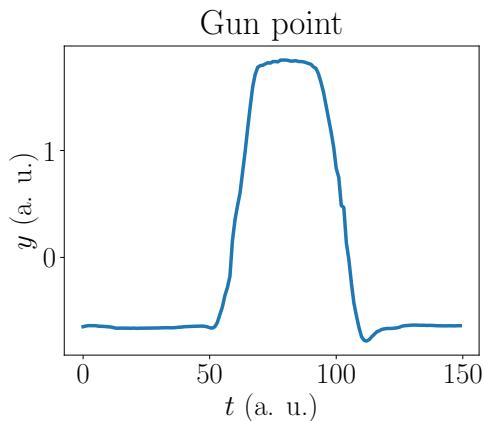
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## Time series data



Time series is some function of time, for example:

$$x = [1, 2, 5, -1, 3]. \quad (1)$$

# Dataset

Consider the following dataset:

- ▶ Train instance from class **a**:  $a = [1, 1, 2, 3, 5]$ .
- ▶ Train instance from class **b**:  $b = [2, 1, 5, 7, 17]$ .
- ▶ Test instance from an unknown class:  $x = [2, 3, 5, 8, 13]$ .

We now apply *altx* to classify the test instance, i.e., determine to which class  $x$  belongs.

## Goal of *altx*

Evaluate how much the data belongs to one class (**closer to 0, more similar!**)

| Data        | Class a | Class b |
|-------------|---------|---------|
| $x_0^{(a)}$ | 0.2     | 3       |
| $x_1^{(a)}$ | 0.03    | 5       |
| $x_2^{(a)}$ | 0.1     | 7       |
| $x_3^{(b)}$ | 4       | 0.1     |
| $x_4^{(b)}$ | 6       | 0.15    |
| $x_5^{(b)}$ | 2       | 0.3     |
| $x^?$       | 0.00035 | 0.029   |

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## Step 1: Extracting the Laws

We define parameters  $r = 3$ ,  $l = 2$ ,  $k = 1$ . (Their meaning will be discussed later.)

From  $a = [1, 1, 2, 3, 5]$  we extract the following segments of length  $r = 3$ :

$$[1, 1, 2], [1, 2, 3], [2, 3, 5]. \quad (2)$$

Then we embed them to  $l \times l = 2 \times 2$  matrices:

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}, \quad \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}. \quad (3)$$

From those, the laws<sup>1</sup> vectors which gives close to 0 when multiplying the embedded matrices, are:

$$\begin{bmatrix} -0.8507 \\ 0.5257 \end{bmatrix}, \quad \begin{bmatrix} -0.8507 \\ 0.5257 \end{bmatrix}, \quad \begin{bmatrix} -0.8507 \\ 0.5257 \end{bmatrix}. \quad (4)$$

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<sup>1</sup>eigenvectors corresponding to the smallest absolute eigenvalues



## Step 1: Extracting the Laws (continued)

We combine the results into one matrix and repeat the process for all training instances (separately for each class).

From class **a**, we obtain laws:

$$L_a = \begin{bmatrix} -0.8507 & -0.8507 & -0.8507 \\ 0.5257 & 0.5257 & 0.5257 \end{bmatrix}. \quad (5)$$

From class **b**, we obtain laws:

$$L_b = \begin{bmatrix} -0.9571 & -0.8702 & -0.9085 \\ 0.2898 & 0.4927 & 0.4179 \end{bmatrix}. \quad (6)$$

## Step 2: Embedding the Test Instance

The test instance  $x = [2, 3, 5, 8, 13]$  is embedded as:

$$E = \begin{bmatrix} 2 & 3 \\ 3 & 5 \\ 5 & 8 \\ 8 & 13 \end{bmatrix}. \quad (7)$$

## Step 3: Projection by Laws

Multiplying with class **a** laws:

$$F_a = EL_a = \begin{bmatrix} -0.0344 & -0.0344 & -0.0344 \\ 0.0132 & 0.0132 & 0.0132 \\ -0.0050 & -0.0050 & -0.0050 \\ 0.0019 & 0.0019 & 0.0019 \end{bmatrix}. \quad (8)$$

Multiplying with class **b** laws:

$$F_b = EL_b = \begin{bmatrix} -0.2898 & -0.0727 & -0.1563 \\ -0.2439 & -0.0252 & -0.1091 \\ -0.2615 & -0.0434 & -0.1272 \\ -0.2548 & -0.0365 & -0.1203 \end{bmatrix}. \quad (9)$$

## Step 4: Feature Calculation

Using the mean of all squared values:

- ▶ Class **a**:  $\text{mean} = 0.00035$ .
- ▶ Class **b**:  $\text{mean} = 0.029$ .

**Conclusion:** The test instance is closer to class **a**.

## Exercise

Using *altx\_intro.ipynb* notebook calculate to which class does

$$x = [3, -1, 5, 3, 13] \tag{10}$$

belongs.

## Exercise – solution

Using *altx\_intro.ipynb* notebook calculate to which class does

$$x = [3, -1, 5, 3, 13] \tag{11}$$

belongs.

Class a score: 0.43, class b score: 0.42  $\Rightarrow$  it belongs to **class b**.

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# Meaning of Parameters $r, l, k$

- ▶  $r$ : Length of the original sequence window used for feature extraction.
- ▶  $l$ : Embedding size, defines a symmetric  $l \times l$  matrix.
- ▶  $2l - 1$ : Number of equally spaced elements sampled from the window of length  $r$ .
- ▶  $k$ : Step size (stride) used for shifting the window over the time series.



## Example $r, l, k$ values

Consider the following instance:

$$x' = [2, 1, 5, 7, 17, 31, 65, 127, 257, 511].$$

Let  $r = 5, l = 2$ , and  $k = 1$ . Then  $2l - 1 = 3$  and

$$x'_1 = [\boxed{2}, 1, \boxed{5}, 7, \boxed{17}], 31, 65, 127, 257, 511].$$

$$x'_2 = [2, \boxed{1}, 5, \boxed{7}, 17, \boxed{31}], 65, 127, 257, 511].$$

$$x'_3 = [2, 1, \boxed{5}, 7, \boxed{17}, 31, \boxed{65}], 127, 257, 511].$$

$$x'_4 = [2, 1, 5, \boxed{7}, 17, \boxed{31}, 65, \boxed{127}], 257, 511].$$

$$x'_5 = [2, 1, 5, 7, \boxed{17}, 31, \boxed{65}, 127, \boxed{257}], 511].$$

$$x'_6 = [2, 1, 5, 7, 17, \boxed{31}, 65, \boxed{127}, 257, \boxed{511}].$$

## Example $r, l, k$ values

Consider the following instance:

$$x'' = [2, 1, 5, 7, 17, 31, 65, 127, 257, 511].$$

Let  $r = 7, l = 2$ , and  $k = 2$ . Then  $2l - 1 = 3$  and

$$x_1'' = [\boxed{2}, 1, 5, \boxed{7}, 17, 31, \boxed{65}], 127, 257, 511].$$

$$x_2'' = [2, 1, \boxed{5}, 7, 17, \boxed{31}, 65, 127, \boxed{257}], 511].$$

## Exercise: Parameters $r, l, k$

**Exercise 1.** Consider  $z = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]$  with parameters  $r = 5$ ,  $l = 2$ ,  $k = 2$ . Mark all windows and the sampled elements inside each window, following the notation of the examples.

**Exercise 2.** The following windows have been identified on  $y = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]$ :

$$y_1 = [\boxed{10}, 20, 30, \boxed{40}, 50, 60, \boxed{70}], 80, 90, 100].$$

$$y_2 = [10, 20, 30, \boxed{40}, 50, 60, \boxed{70}, 80, 90, \boxed{100}].$$

Determine the parameters  $r$ ,  $l$ , and  $k$ .

## Exercise: Parameters $r, l, k$ – Solutions

**Exercise 1.** With  $r = 5$ ,  $l = 2$ ,  $k = 2$ : sampling step  $= \frac{r-1}{2l-2} = 2$ , so  $2l - 1 = 3$  elements are sampled per window.

$$z_1 = [\boxed{1}, 2, \boxed{3}, 4, \boxed{5}, 6, 7, 8, 9, 10].$$

$$z_2 = [1, 2, \boxed{3}, 4, \boxed{5}, 6, \boxed{7}, 8, 9, 10].$$

$$z_3 = [1, 2, 3, 4, \boxed{5}, 6, \boxed{7}, 8, \boxed{9}, 10].$$

**Exercise 2.**

- ▶ Each window spans 7 elements  $\Rightarrow r = 7$ .
- ▶ Each window contains  $2l - 1 = 3$  sampled elements  $\Rightarrow l = 2$ .
- ▶ Windows start 3 positions apart  $\Rightarrow k = 3$ .

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# Anomaly Detection as a Special Case of Classification

Anomaly detection is classification with **one class**: the normal class.

- ▶ Laws are extracted from **normal** data only.
- ▶ Instead of one scalar per class, we compute one score **per embedded window**, giving a score *vector*.
- ▶ **Peaks** in the score vector mark the anomaly locations.

## Step 1: Laws from Normal Data

Normal data:  $a = [1, 1, 2, 3, 5, 8, 13]$ , parameters  
 $r = 3$ ,  $l = 2$ ,  $k = 1$ .

$$L = \begin{bmatrix} -0.8507 & -0.8507 & -0.8507 & -0.8507 & -0.8507 \\ 0.5257 & 0.5257 & 0.5257 & 0.5257 & 0.5257 \end{bmatrix}. \quad (12)$$

All five laws are identical because every window follows the same Fibonacci rule.

## Step 2: Embedding the Test Instance

Test instance (anomalies at positions 0 and 3):

$x = [6, 1, 2, 8, 5, 8, 13]$ .

$$E = \begin{bmatrix} 6 & 1 \\ 1 & 2 \\ 2 & 8 \\ 8 & 5 \\ 5 & 8 \\ 8 & 13 \end{bmatrix}. \quad (13)$$

Row  $j$  is the pair  $(x_j, x_{j+1})$ .



## Step 3: Projection

$$F = E \cdot L = \begin{bmatrix} -0.753 & -0.753 & -0.753 & -0.753 & -0.753 \\ 0.090 & 0.090 & 0.090 & 0.090 & 0.090 \\ 0.304 & 0.304 & 0.304 & 0.304 & 0.304 \\ -0.443 & -0.443 & -0.443 & -0.443 & -0.443 \\ -0.005 & -0.005 & -0.005 & -0.005 & -0.005 \\ 0.002 & 0.002 & 0.002 & 0.002 & 0.002 \end{bmatrix}. \quad (14)$$

Rows 0 and 3 have the largest absolute values.

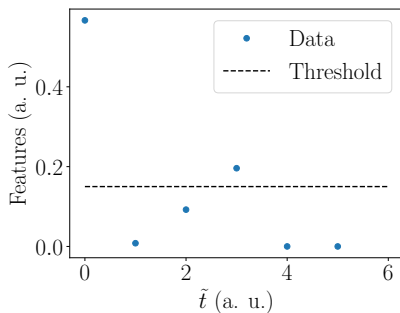
## Step 4: Anomaly Scores

Average  $F^2$  over the laws (columns), keeping one score per row:

$$s_j = \text{mean}_m(F_{jm}^2). \quad (15)$$

$$s = [\underbrace{0.567}_{\text{anomaly}}, 0.008, 0.092, \underbrace{0.196}_{\text{anomaly}}, 0.000, 0.000]. \quad (16)$$

Threshold  $\tau = 0.15$ : gives anomalies (peaks) at 0 and 3.



The mean can be replaced by a **percentile** (e.g. 5<sup>th</sup>) for robustness.

## Index Mapping: Score Index to Original Position

The score vector has  $N - l + 1$  elements. Score  $s_j$  is computed from window

$$E_j = [x_j, x_{j+1}, \dots, x_{j+l-1}], \quad (17)$$

so a peak at index  $j$  indicates an anomaly at original position  $j$ .  
Notebook example ( $l = 2$ ):

- ▶  $s_0 = 0.567 > \tau \Rightarrow$  anomaly at position 0. ✓
- ▶  $s_3 = 0.196 > \tau \Rightarrow$  anomaly at position 3. ✓

## Exercise: Index Mapping

$altx$  is applied to a time series of length  $N = 10$  with  $l = 3$ .

**Part (a).** How many elements does the anomaly score vector  $s$  have?

**Part (b).** The score vector has peaks at indices  $j = 1$  and  $j = 6$ . What are the corresponding anomaly positions in  $x$ ?

## Exercise: Index Mapping – Solutions

**Part (a).**

$$N - l + 1 = 10 - 3 + 1 = 8. \quad (18)$$

**Part (b).** A peak at index  $j$  indicates an anomaly at position  $j$ :

- ▶ Peak at  $j = 1 \Rightarrow$  anomaly at position 1.
- ▶ Peak at  $j = 6 \Rightarrow$  anomaly at position 6.

# Summary of ALT Method

- ▶ Extract subsequences and embed into symmetric matrices.
- ▶ Compute eigenvectors of smallest absolute eigenvalues — these are the laws.
- ▶ Multiply embedded test instances with class-specific laws.
- ▶ Compute summary statistics (e.g., mean of squared values).
- ▶ Classify by choosing the class that minimizes the statistic.
- ▶ By only taking statistic over the laws, the peaks represent anomalies.